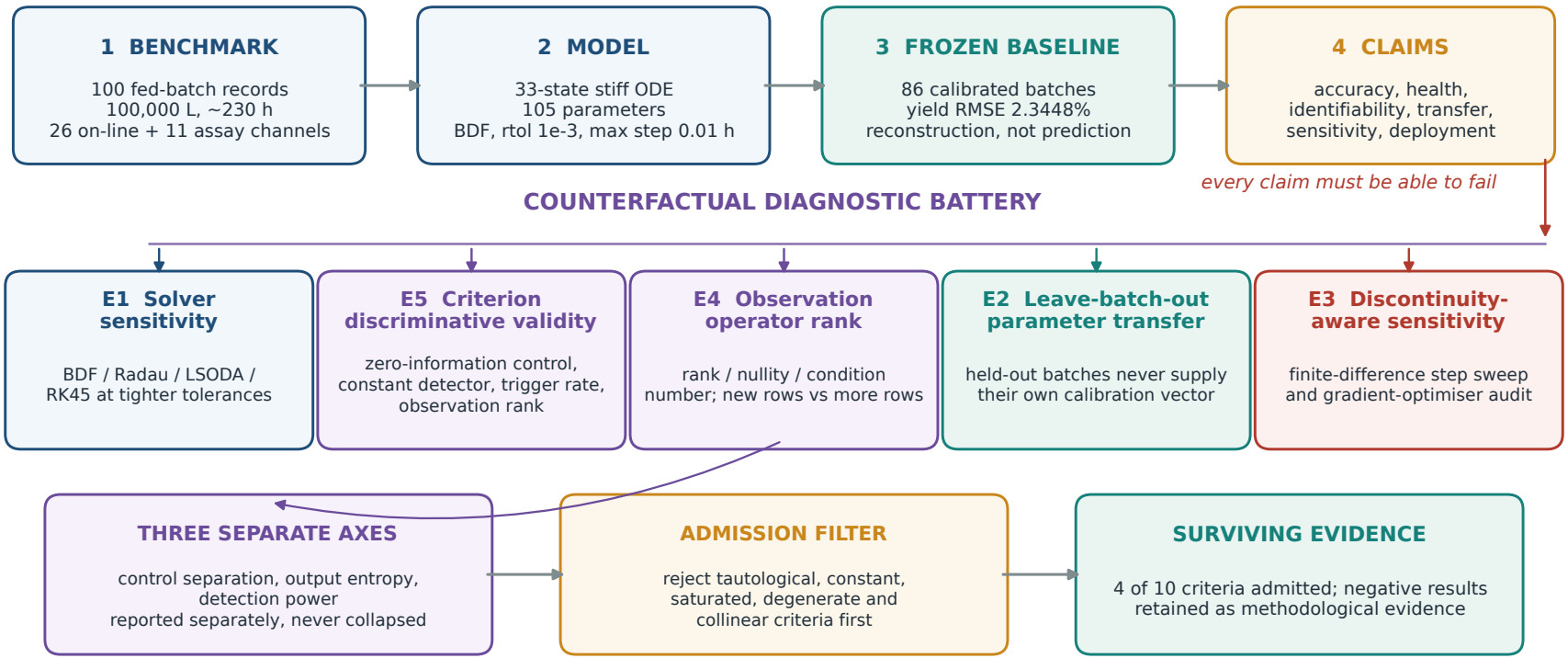
Counterfactual validation protocol for process simulation models

IndPenSim penicillin fed-batch benchmark: industrial-scale demonstration, not the contribution



All experiments are in silico; no wet-laboratory or prospective plant data are introduced.